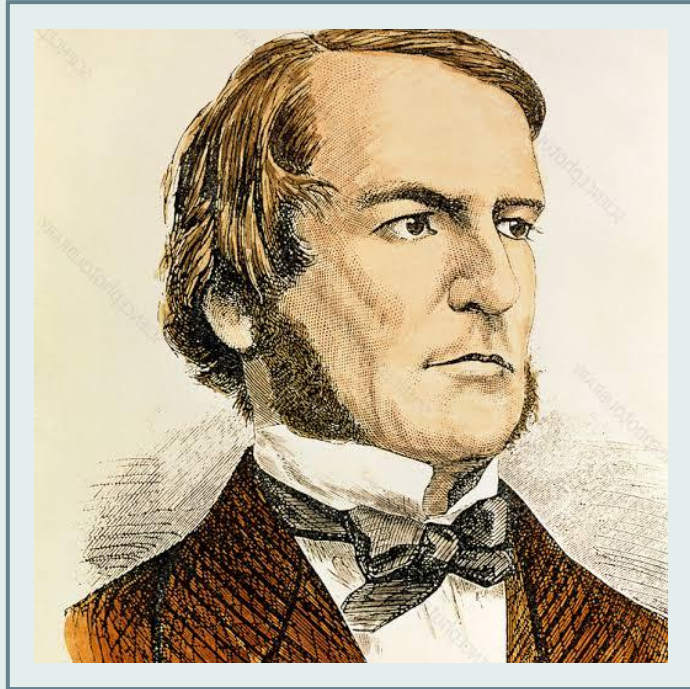


George Boole

1815 - 1864

by Fausto Nunes



Who was George Boole?



English mathematician who helped establish modern **symbolic logic** and whose algebra of logic, now called **Boolean algebra**, is essential to the design of digital computer circuits.



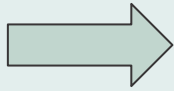


AND a lot of other things:

- Differential equations
- Boolean data type, function, expression, network, circuit, processor, etc.



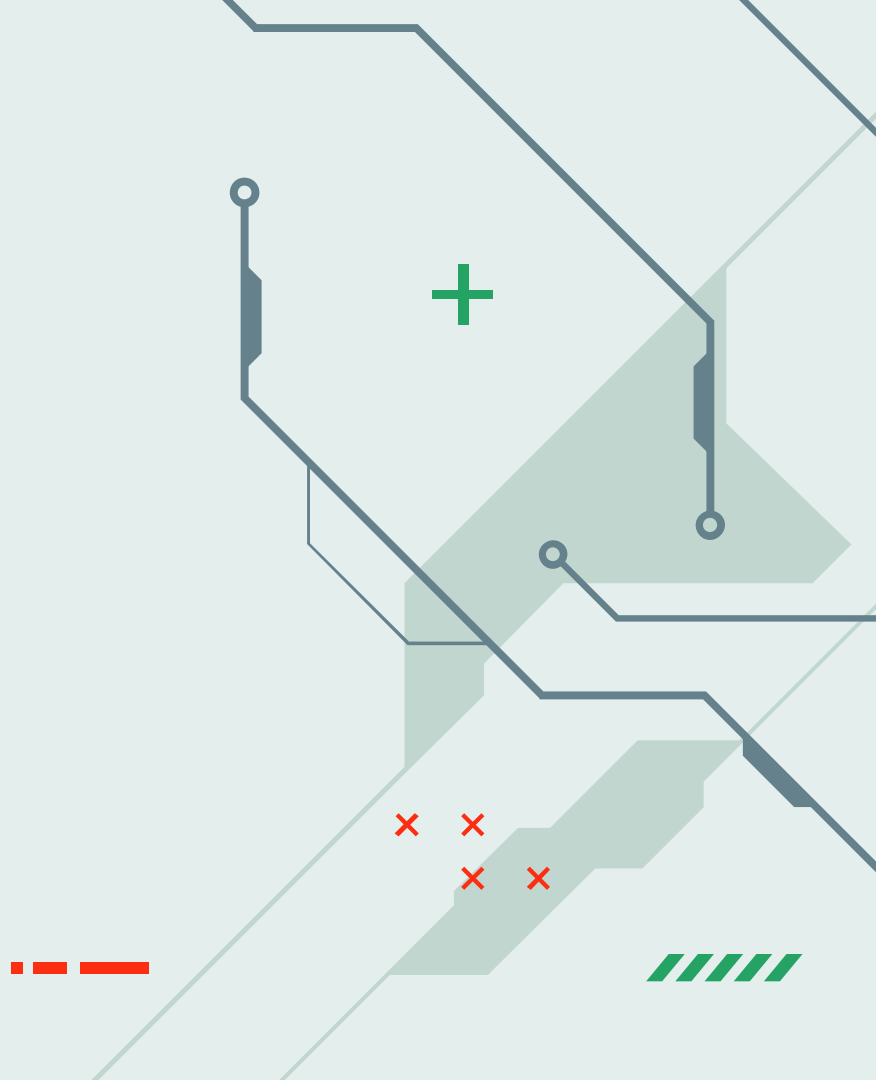
0 1



Thanks Boole!

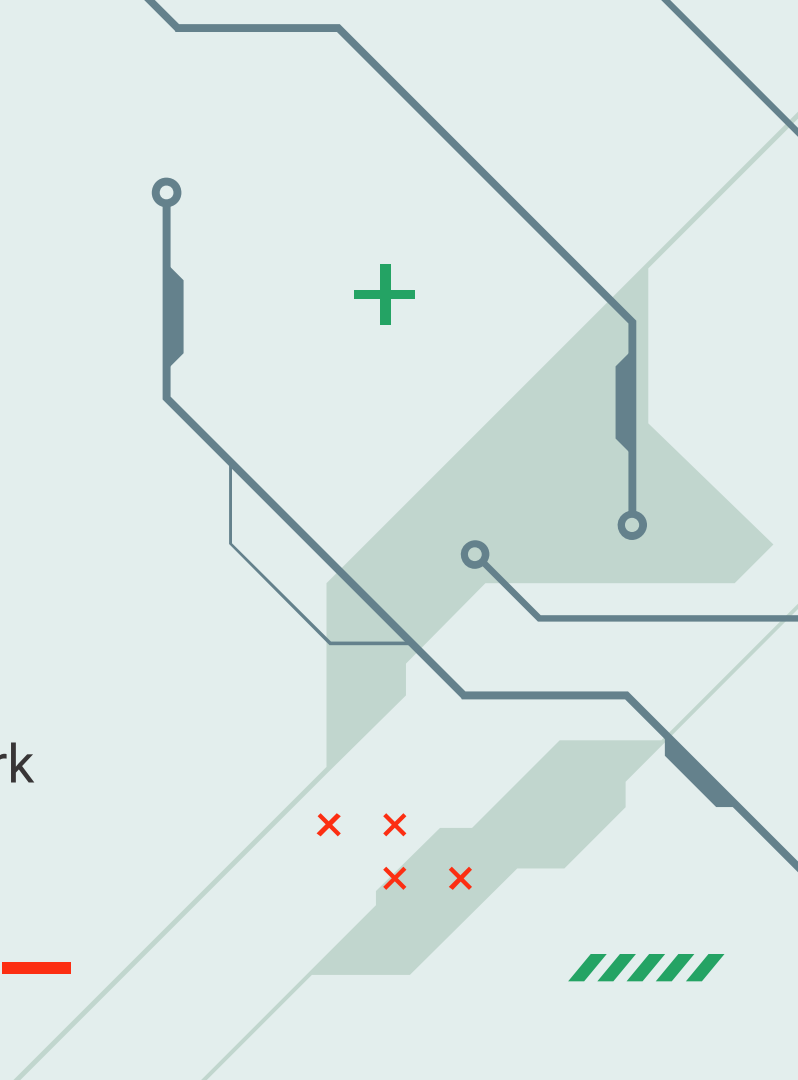
Some life facts:

- Family of very modest means
- Self-taught



Some life facts 2:

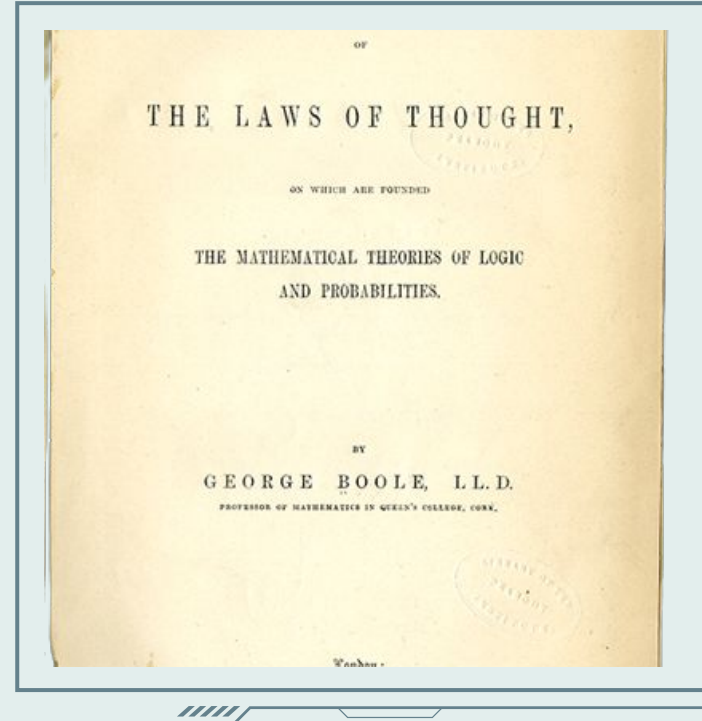
- Teaching assistant at 16
- Founded his own school at 19
- Became the first professor of mathematics at Queen's College, Cork



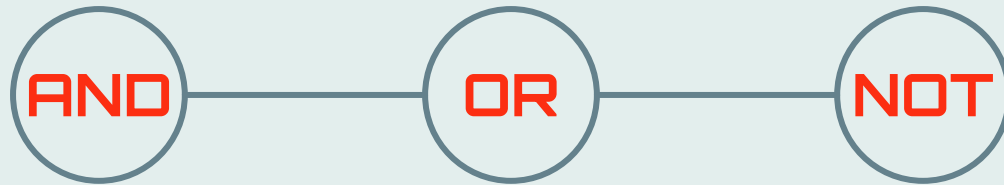


PUBLISHED WORK

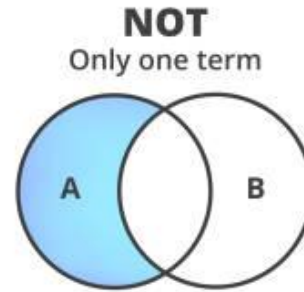
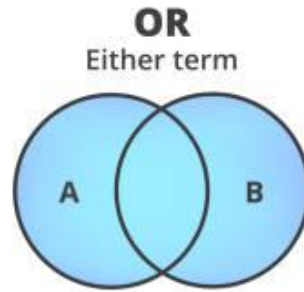
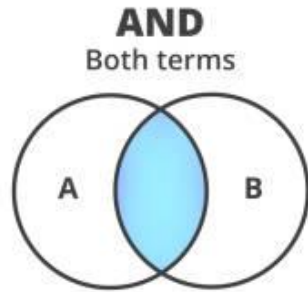
- The Mathematical Analysis of Logic (1847)
- Laws of Thought (1854)



Boolean Logic Operands



BOOLEAN LOGIC



x x
x x

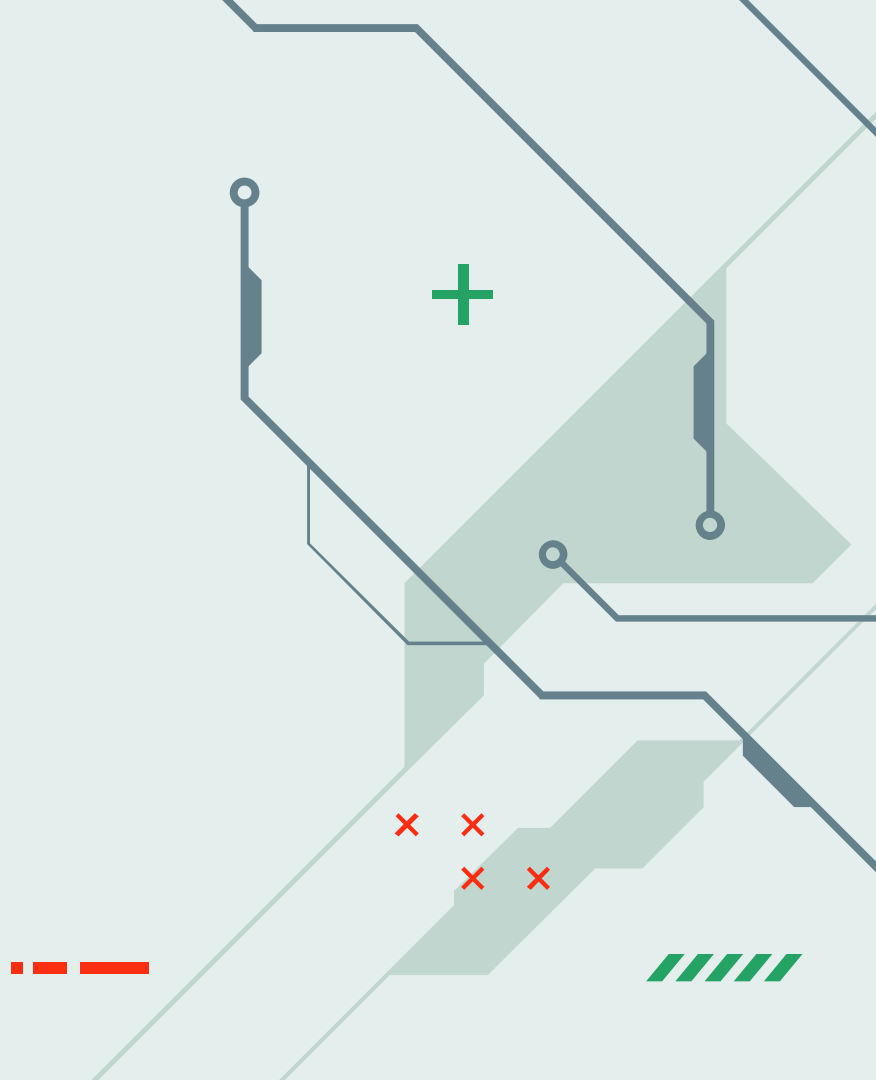
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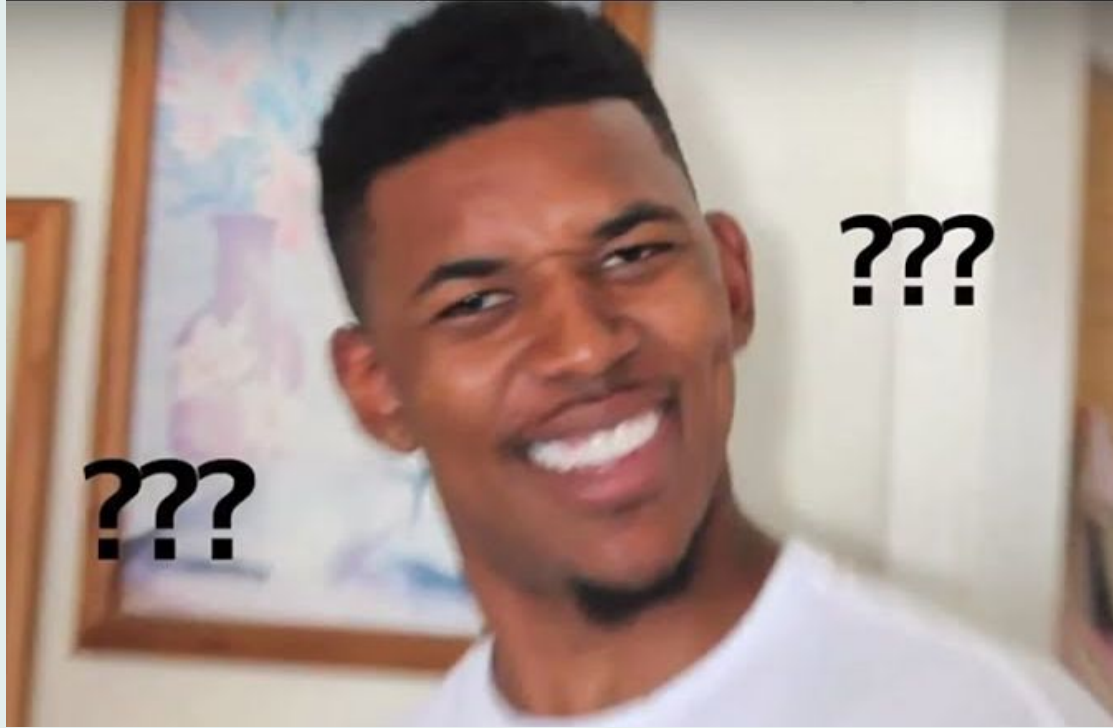
x	y	x AND y	x OR y	NOT x	NOT y
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	0	1
1	1	1	1	0	0

+

x x
x x

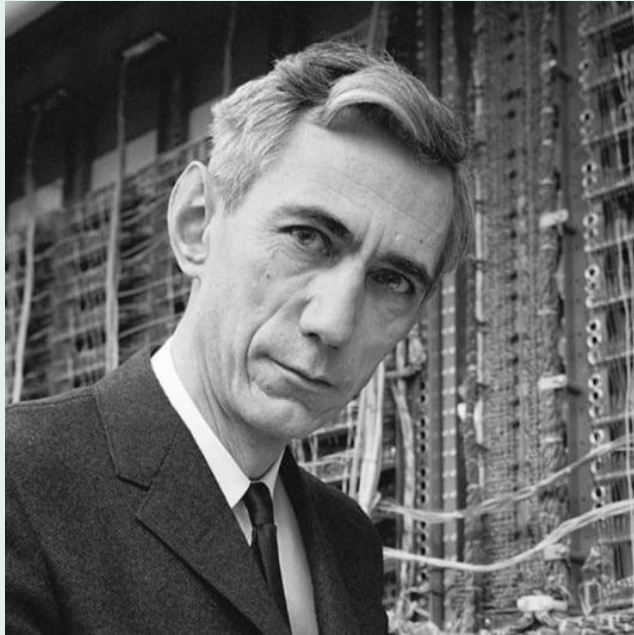
Okay cool... but,
how does his
work relate to
computers?





+

x x
x x



- In 1937, **Claude Shannon** wrote a master's thesis on the applicability of Boolean logic to certain relay switching systems
- True (1) and false (0) - Open or Closed circuit

Since then, Boolean logic and computation evolved gradually into what we use today!



The slide features decorative geometric patterns in the corners. The top-left corner has a series of nested, stepped lines in shades of green and blue, with a red plus sign (+) and five green 'x' marks. The bottom-right corner has similar stepped lines, a red circle, and five green 'x' marks. The central text is in a bold, dark grey, sans-serif font.

Thank you!
Any questions?